Xi Lin

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Research Interests

My research interest is in **World Action Models (WAMs)** and embodied foundation models: learning action-conditioned world representations that preserve task-relevant physical consequences. I focus on scalable robot action data, dexterous manipulation, counterfactual future evaluation, and control-sufficient representations for reliable robot decision-making.

Education

Johns Hopkins University — Baltimore, MD
M.S. in Robotics — *Expected Jun. 2027*

Dalian University of Technology — Dalian, China
B.S. in Mechanical Engineering (Joint Program with UC Irvine); Dual Degree in Automation — *Jun. 2025*

- **Thesis:** *Robot Design and Reinforcement Learning Control for Complex Terrain*; **Honors:** National 3D Design Competition (3rd Prize), DUT Sci-Tech Innovation Scholarship

Selected Publications & Patents

IROS 2026 (Under Review | 1st Author): *MORN: Metacognitive Object-Goal Regulation for Resource-Rational Long-Horizon Navigation*
ECCV 2026 (Under Review | 3rd Author): *Dual-Anchoring: Addressing State Drift in Vision-Language Navigation*
IEEE IPIC 2023 (1st Author): *Research on Object Detection of Robotic based on Convolutional Neural Network*
Patents: Spiral Closed Anti-blocking Conveyor and Grinding Device (1st Inventor); Robotic Arm (2nd Inventor)

Selected Research Experience

JD.com Explore Academy — Beijing, China
Embodied Decision-Making Research Intern — *Nov. 2025 – Mar. 2026*

Metacognitive Regulation for Long-Horizon Embodied Search (MORN | 1st Author Under Review)

- Designed a metacognitive regulation framework that evaluates when an embodied agent should continue, switch, backtrack, or terminate under partial observability and long-horizon uncertainty.
- Decoupled low-level robot execution from high-level action/goal consequence evaluation, enabling resource-aware embodied decision-making beyond greedy visual-language matching.
- Built a zero-shot multi-goal benchmark on HM3D and validated the system in simulation and on real robots, improving Goal Completion Rate by 33% and reducing Wasted Steps by 22% over baselines.

World-Model Supervision for Embodied State Anchoring (Dual-Anchoring | 3rd Author Under Review)

- Constructed supervised training tasks for world-model-based state anchoring, helping the agent preserve task progress, visual history, and landmark-centric state representations over long horizons.
- Supported data collection, curation, model training, evaluation, and edge deployment for a real-robot embodied navigation pipeline in large-scale commercial scenarios.

Xiaomi Robotics Department — Beijing, China
Humanoid Robot Learning Intern — *Sep. 2025 – Nov. 2025*

- Converted factory assembly videos into SMPL trajectories and retargeted human motions to full-size humanoids with PHC, creating reusable action data for downstream robot learning.
- Built high-fidelity humanoid simulation environments in Unitree RL Gym and trained low-level tracking policies for dexterous assembly-style robot behaviors.
- Benchmarked ASAP and related pipelines in a unified physics stack across payload walking, tightening, and manipulation-like work conditions, producing automated evaluation tools for hardware-policy iteration.

Institute for AI Industry Research, Tsinghua University — Wuxi, China
Embodied AI Research Intern (Undergraduate Thesis) — *Jan. 2025 – Aug. 2025*

- Built procedural terrain environments in Isaac Gym/HPL and designed curriculum-based RL training for Unitree H1 locomotion on unstructured terrain.
- Integrated ROS2 control interfaces, SLAM, and 3D semantic perception into a real-robot system for mapping, obstacle avoidance, and sim-to-real locomotion deployment.

Johns Hopkins University Terradynamics Lab Baltimore, MD
Research Assistant *Jun. 2024 – Dec. 2024*

- Built and optimized an LSTM-based dynamics model for trajectory anticipation in cockroach-inspired locomotion, achieving a Spearman correlation coefficient of 0.87.
- Developed a GMM-based trajectory data generator that expanded the effective training set for complex interaction scenarios by up to 300%.
- Derived non-holonomic constraints and quantified the nonlinear relationship between angular velocity and pose error to support physically grounded control analysis.

Dalian University of Technology Dalian, China
Undergraduate Researcher *Oct. 2022 – Jun. 2023*

- Improved YOLOv5 with adaptive convolution, anchor adjustment, and multi-scale training, then integrated the detector with VINS-Fusion for real-time UAV perception and navigation.
- Designed and prototyped patented mechanical devices in SolidWorks, building early experience in robot hardware, CAD modeling, and physical system iteration.

Technical Skills

World Models & Embodied AI: World Action Models, Action-Conditioned Prediction, VLM/VLA, Object-Centric and State-Anchored Representations
Robot Learning & Action Data: Reinforcement Learning, Motion Retargeting, Humanoid/Dexterous Action Data, Sim-to-Real, Whole-Body Control
Robotics Systems: ROS/ROS2, SLAM, Open-Vocabulary Perception, 3D Vision, Real-Robot Deployment
Programming & ML: Python, PyTorch, OpenCV, C++, MATLAB
Simulation & Tools: Isaac Gym/Lab, Habitat, MuJoCo, Unitree RL Gym, Docker, Git, Linux, SolidWorks